ARCHITECTURE SPECIFICATION // IBM ENTERPRISE DATATHON

RED STRING PLATFORM OVERHAUL

AIR-GAPPED TELEMETRY ENGINE & NON-INVASIVE EDGE INTRUSION DETECTION

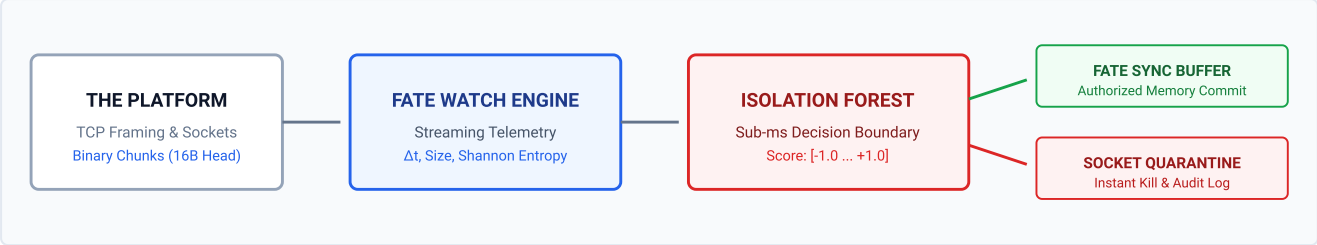
Date: October 2026
Target: LinuxONE / Edge Enclaves
Status: Validated Prototype

< 1.4 ms	0 B Cloud	O(1)	100%
EDGE INFERENCE LATENCY	EXTERNAL DATA LEAKAGE	ENTROPY VECTOR EXTRACTION	ZERO-TRUST LOCAL AUDIT

1. EXECUTIVE PARADIGM & DATATHON RE-ARCHITECTURE

REFACTORING OBJECTIVE

Enterprise environments strictly prohibit uninspected P2P data transports due to the risk of covert exfiltration channels and lateral insider threats. Red String is transformed from a passive socket synchronizer into an active, **Zero-Trust Behavioral Telemetry Fabric**. Instead of inspecting or decrypting encrypted customer payloads, Red String inspects transmission kinetics via *Fate Watch*: chunk intervals, byte-level entropy, and sliding-window frame frequencies.



2. BINARY CHUNK FRAMING SPECIFICATION

ZERO-OVERHEAD WIRE PROTOCOL

BYTE OFFSET	FIELD NAME	DATA TYPE	ENTERPRISE FUNCTION
`0x00 - 0x03`	**Magic Header**	4-Byte ASCII	Identifies verified protocol cluster (`RDSG`) without cloud handshake.
`0x04`	**Channel ID**	1-Byte Int	Flags stream target: Clipboard (0x01), Keystroke (0x02), File (0x03).
`0x05 - 0x0C`	**Epoch Timestamp**	8-Byte Big-Endian	Provides millisecond timestamp for inter-arrival drift analytics.
`0x0D - 0x10`	**Payload Length (N)**	4-Byte Unsigned Int	Fixed boundary enforcement prevents socket buffer starvation.
`0x11 - 0x11+N`	**Raw Content**	N Bytes (Opaque)	Synchronized payload consumed only if anomaly scoring passes.

3. REAL-TIME TELEMETRY & ANOMALY ENGINE

NON-INVASIVE MATHEMATICAL PROFILE

Feature Vector Extraction Formulation

For each discrete frame arriving at socket boundary `i`, calculate the 4-dimensional statistical signature in sub-millisecond time without payload decryption:

1. Inter-arrival Gap: $\Delta t = t_i - t_{i-1}$ (detects automated bot loops)

2. Shannon Entropy: $H(X) = -\sum P(x) \cdot \log_2 P(x)$ (distinguishes text vs. dumps)

3. Frame Size: $S_i = \text{len}(\text{Payload})$ (identifies mass file leaks)

4. Sliding Burst Ratio: $B_{5s} = \text{Count}(\text{packets}) / 5000\text{ms}$

4. COMPREHENSIVE DEPENDENCY & PLUGIN MANIFEST

PRODUCTION PYTHON TOOLCHAIN

PACKAGE / PLUGIN	VERSION TARGET	DATATHON IMPLEMENTATION ROLE
`scikit-learn`	`>= 1.3.0`	Trains unsupervised `IsolationForest` on baseline human sync dynamics.
`scipy`	`>= 1.11.0`	Computes streaming Shannon entropy across the 256-byte frequency distribution.
`streamlit & plotly`	`Latest Stable`	Renders real-time operations console: live score dial, burst scatter, kill toggle.
`opencv-python`	`headless >= 4.8`	High-speed QR parsing for mutual visual authentication on air-gapped nodes.
`qrcode[pil]`	`>= 7.4.2`	Generates ephemeral, time-keyed "Magic Circle" pairing tokens for direct handshakes.
`joblib`	`>= 1.3.0`	Serializes and cold-boots the pre-fitted contamination boundary model in < 20ms.

5. Enterprise Datathon Live Demonstration Script (24-Hour Pitch)

Phase I: Baseline Operations (0:00 - 1:30): Demonstrate normal cross-device clipboard sync between workstation and edge target via binary TCP framing. The Streamlit console shows green status markers with nominal anomaly scores (`Score > 0.15`, Entropy ~4.1 for plain text).

Phase II: Threat Ingestion & Automated Quarantine (1:30 - 3:00): Execute an automated script simulating malicious exfiltration (e.g., 2,500 payload bursts at 2ms intervals, high-entropy zipped database extract). The telemetry engine catches the deviation in < 2 frames, drops the receiving buffer before memory commit, triggers a red alert on the dashboard, and outputs a tamper-proof forensic audit trail.

6. DEPLOYMENT COMMAND

QUICKSTART EXECUTION

```
pip install scikit-learn scipy numpy pandas streamlit plotly qrcode[pil] opencv-python joblib
```